

□ 24 ANOVA assumptions

ANOVA revisited

Recall that ANOVA is a statistical method to allow us to decide if there's a difference in a scale variable at the 3 or more levels of a nominal variables (or equivalently, in 3 or more samples)

For example, we use ANOVA to decide if the amount spend by customers differed depending on the whether they bought electronics, jewellery or fragrances.

Or to decide if say the average calorie content of an animal feed differs by raw material supplier - Great Grains (G), Grain Chain (C) or Kelly's Food (K) .

The following assumptions for ANOVA must be satisfied

- The sample data is independent (as with all tests)
- The data in each sample (i.e. goods type, bar manufacturer etc.) should be normally distributed
- The variance (i.e. the spread) in each sample should be the same (homoscedastic)

We have looked at the first two, here we consider the third.

Homoscedasticity

A dot plot of the data for the animal feed example is shown opposite.

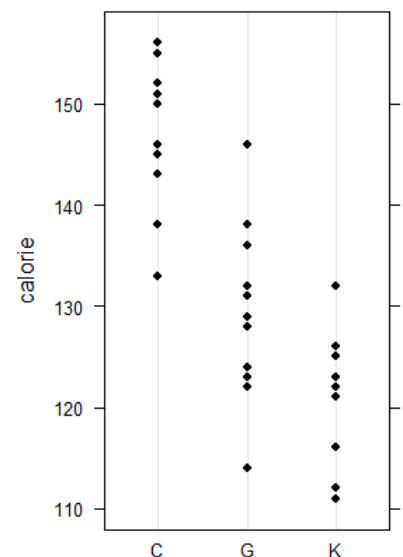
For all three raw material suppliers, there is clearly variation, or spread in the calorific values in the sample from that supplier.

We need to establish that the spread is about the same for all three – i.e. that the variance is homoscedastic.

Comparing standard deviations is a useful first step.

```
tapply(calorie, grain, sd)
      C      G      K
7.315405 8.490190 6.468455
```

The values seem fairly similar and this is consistent with the dot plot.
But we would like a formal hypothesis test to establish this objectively.
One such test is Levene's test,



Packages in R

We digress for a moment to discuss R packages. R packages are a mechanism to organise your work and, if you wish, to share it. Typically, packages include code, data and documentation.

Sometimes useful constructs that may not be available in the basic R installation, will likely have been developed by someone else and made available to the R community.

To make use of such a package you must, via the **Packages** menu in R, **Set CRAN mirror**, **Install** the package and then **Load** the package.

An implementation of Levene's test is available in a package called *lawstat*. The data must be in stacked form – i.e. one scale variable containing all the *calorie* values and one nominal variable indicating the *grain* supplier.

H_0 : The variance is the same for all raw material suppliers

H_A : The variance is not the same for all raw material suppliers

```
> levene.test(calorie, grain)

      modified robust Brown-Forsythe Levene-type
      test based on the absolute deviations from
      the median

data:  calorie
Test Statistic = 0.29161, p-value = 0.749
```

As $p > 0.05$ we can assume that the variances are the same in all three groups.

What about normality? Is calorific content normally distributed for each supplier?

As the data is stacked, we need the following calls to `shapiro.test`

```
> shapiro.test(calorie[grain=="C"])      # p = 0.4719
> shapiro.test(calorie[grain=="G"])      # p = 0.9701
> shapiro.test(calorie[grain=="K"])      # p = 0.5815
```

So, the assumption of normality is satisfied as is the assumption of edasticity, so we can report our statistically significant finding.

Example

Check the assumptions for the ANOVA that established that customers spend different mean amounts depending on whether they bought electronics, jewellery or fragrances.

Normality?

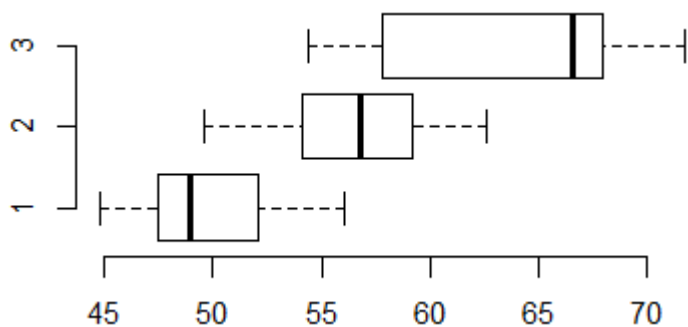
```
> shapiro.test(q4_spend[q3_goods==1])    # p = 0.25
> shapiro.test(q4_spend[q3_goods==2])    # p = 0.54
> shapiro.test(q4_spend[q3_goods==3])    # p = 0.10
```

So, the normality assumption is satisfied,

Homoscedasticity?

```
> tapply(q4_spend, q3_goods, sd, na.rm=T)
      1      2      3
3.408604 3.624857 6.094684

> boxplot(q4_spend~q3_goods, horizontal =T) # Alternative to the dot plot
```



The variance for the third group (fragrances) does appear to be a little greater

```
> levene.test(q4_spend, q3_goods)

      modified robust Brown-Forsythe Levene-type test based on the
      absolute deviations from the median

data:  q4_spend
Test Statistic = 6.3008, p-value = 0.01812
```

As the p value is < 0.05 we must reject the null hypothesis that the variances are equal.

And so, in this example, the assumptions of the test are not all satisfied.

We shouldn't really report a finding. So, what do we do?

Some argue that ANOVA is robust to departures from both normality and homoscedasticity and would likely be unconcerned about the violation above. And proceed to report the finding.

A more cautious approach, and the one we will take, is to abide rigidly by the findings of these tests. In which case, we cannot report a statistically significant finding and might opt to use an alternative, non-parametric test.

□ Non parametric tests

The t test (paired and independent samples) and ANOVA all make assumptions about the data being normal. In addition, ANOVA makes an assumption about homoscedasticity.

If any of these assumptions are violated you should use the non-parametric equivalent. We will not consider these tests in detail in this module – they are listed below for your information.

Parametric test	Nonparametric equivalent
<code>t.test(..., paired = FALSE)</code>	Mann Whitney test
<code>t.test(..., paired = TRUE)</code>	Wilcoxon test
<code>aov()</code>	Kruskal-Wallis test

Exercise

Identify the functions in R that perform these non-parametric tests.